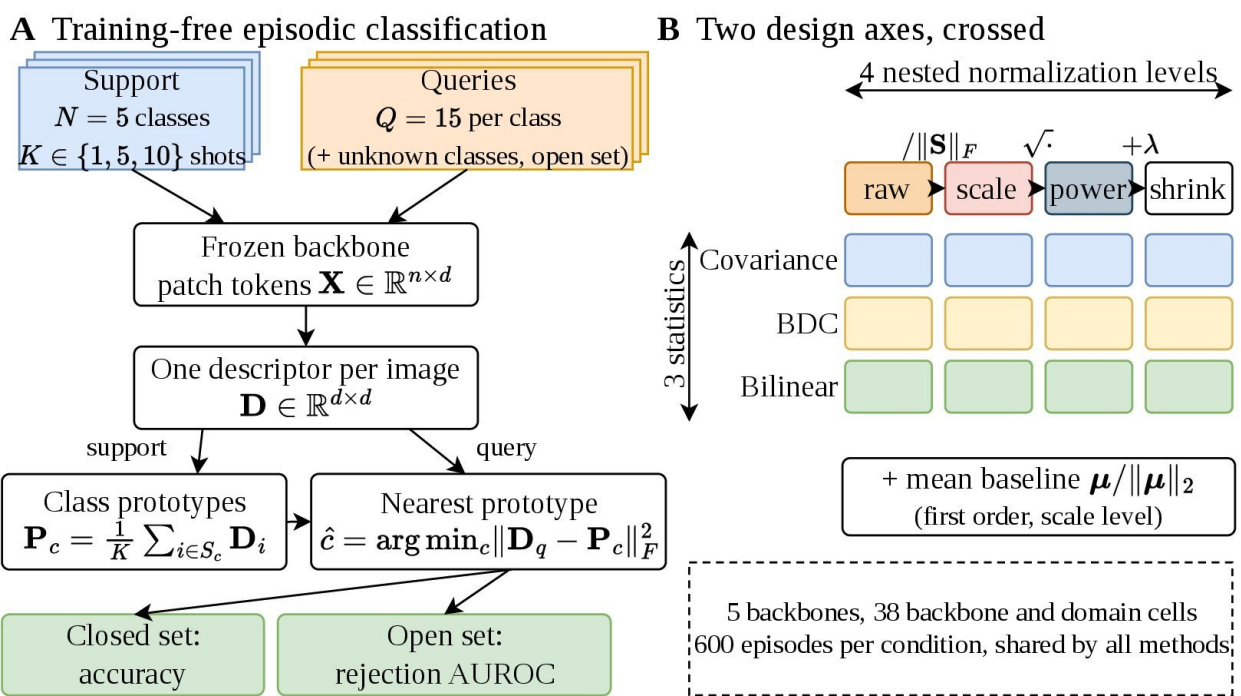

A Training-free episodic classification
Support
$N = 5$ classes
$K \in \{1, 5, 10\}$ shots
Queries
$Q = 15$ per class
(+ unknown classes, open set)
Frozen backbone
patch tokens $\mathbf{X} \in \mathbb{R}^{n \times d}$
One descriptor per image
$\mathbf{D} \in \mathbb{R}^{d \times d}$
support
query
Class prototypes
$\mathbf{P}_c = \frac{1}{K} \sum_{i \in S_c} \mathbf{D}_i$
Nearest prototype
$\hat{c} = \arg\min_c \|\mathbf{D}_q - \mathbf{P}_c\|_F^2$
Closed set:
accuracy
Open set:
rejection AUROC
B Two design axes, crossed
4 nested normalization levels
$/\|\mathbf{S}\|_F$
$\sqrt{\cdot}$
$+\lambda$
raw
scale
power
shrink
3 statistics
Covariance
BDC
Bilinear
+ mean baseline $\boldsymbol{\mu}/\|\boldsymbol{\mu}\|_2$
(first order, scale level)
5 backbones, 38 backbone and domain cells
600 episodes per condition, shared by all methods